

Controlling Embedding Spaces with Text-Conditioned Transformations

Joseph Fiorese^{1*}, Fabian Caba Heilbron², Pankaj Nathani²,
Mubarak Shah¹, and Kushal Kafle²

¹ Institute of Artificial Intelligence, University of Central Florida, USA

² Adobe Research, USA

joseph.fioresi@ucf.edu, {caba, pankaj, kkafle}@adobe.com, shah@crcv.ucf.edu
https://joefioresi718.github.io/ControlEmbed_webpage/

Abstract. Multimodal embedding spaces in models like CLIP enable powerful capabilities such as semantic similarity retrieval and cross-modal zero-shot classification. These embeddings compress high-level semantics into a single vector, which comes at the cost of primarily expressing a dominant semantics like main object while suppressing other important attributes such as camera angle or color tone. We propose a text-conditioned transformation of visual embeddings that makes such attributes explicitly accessible. Given a natural language description of an attribute category (e.g., “color” or “art style”), a network generates an affine transformation that emphasizes the specified attribute. Conditioning on text enables it to learn many attributes simultaneously, accessing them at inference time through an intuitive interface. The network is trained to align transformed embeddings with the frozen latent space, enabling retrieval using existing large-scale embeddings without any re-encoding. When applied to a full set, the same mechanism transforms the latent space for attribute disentanglement tasks such as multi-clustering. By operating directly in latent space, our method provides a unified and efficient framework for controlling embedding spaces, demonstrating state-of-the-art performance across both attribute-based retrieval and multi-attribute organization tasks with near-zero inference cost.

Keywords: Vision-Language Models · Embedding Spaces

1 Introduction

Large-scale vision-language models like CLIP [39] and SigLIP [66] have become ubiquitous in computer vision, learning powerful multimodal embeddings from web-scale data. Such embeddings enable zero-shot task transfer, cross-modal retrieval, and semantic similarity search, and are increasingly deployed in practice. However, their versatility comes from compressing diverse visual concepts—objects, attributes, styles, and contexts—into a single high-dimensional space.

* Majority of work done as an intern at Adobe Research

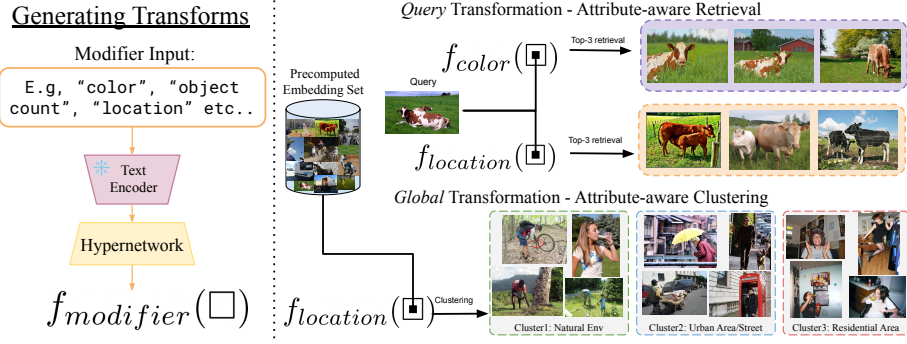


Fig. 1: Overview of our text-conditioned generation of embedding-space transformations. (Left) We propose a hypernetwork that maps natural-language attribute descriptions to a modification function. Applying this transformation to a query embedding produces an attribute-aware representation that emphasizes the requested property. (Right) By controlling the embedding space with attribute transforms, the same mechanism enables both attribute-aware retrieval and attribute-specific clustering.

As a result, important attributes such as color, camera angle, or mood are encoded only implicitly, entangled with more dominant object-level semantics. This entanglement limits the kinds of similarity queries that can be expressed. When retrieving images from a reference image with CLIP, we can neither predict nor control whether the similarity will be based on the main subject, style, or other attributes. For example, suppose a reference image is a pastel-colored watercolor painting of a Corgi as viewed from a low angle. In this case, CLIP can retrieve images that look similar, but the specific properties are nebulous [46]. Will a photograph of a Corgi against a plain background rank higher or lower than a similarly styled painting of a Labrador viewed from a similar angle? Not only can we not readily know the answer, but we also cannot easily specify our preference.

This limitation restricts the feasibility of *attribute-aware* retrieval, where users search based on specific attributes rather than broad semantic categories. For instance, given a large collection of photos, one user might want to retrieve images with similar composition or lighting, while another might care more about color or artistic style. Yet, because such attributes are entangled within the latent space, a single query embedding cannot easily emphasize them. This challenge extends beyond retrieval to other tasks such as *multi-clustering*, where different users may wish to group the same media collection by different attributes. Together, these limitations highlight the need for controllable transformations of embedding spaces that adapt dynamically to user-specified attributes.

Existing methods for attribute disentanglement often rely on one of two major strategies: (1) finetuning the encoder to explicitly separate desired attributes [10, 53, 64, 65]; or (2) scaling up inference-time requirements by composing knowledge from large VLMs and LLMs [21, 26, 31]. The first approach enables efficient inference once trained, but demands costly finetuning and re-embedding. This causes linear memory footprint scaling and sacrifices the generalization of

the original model. The second requires no retraining, but introduces high inference latency and limited scalability for real-time applications.

In this paper, we propose a novel approach to address the above limitations. At the core of our approach is a simple but powerful idea: instead of retraining the encoder or designing separate attribute-specific heads, we learn to reshape the latent space itself through a text-conditioned transformation. Concretely, we train an MLP-based hypernetwork that takes as input a text embedding representing an attribute category (e.g., “color” or “style”) and outputs a single global affine transformation. This transform acts as a learned operator that modifies the existing embedding space so that the specified attribute becomes more salient. Figure 1 demonstrates this idea, showcasing its use in two complementary applications: a *query* transformation, where only the query embedding is modified to emphasize a desired attribute for attribute-based retrieval from a static embedding index; and a *global* transformation, which transforms all embeddings for downstream analysis such as unsupervised clustering. Because the encoder remains frozen and the hypernetwork operates directly on latent embeddings, inference is near-instant and the method can be seamlessly integrated into existing multimodal pipelines.

Our main contributions are summarized as follows:

- We introduce a **text-conditioned hypernetwork** that generates transformations of visual embeddings, enabling controlled emphasis of diverse attributes such as color, mood, and camera angle. Alignment with base embeddings enables a *query* mode for attribute-aware retrieval.
- These transformations can be applied to disentangle entire embedding sets in a *global* setting. This induces an **attribute-specific similarity metric**, where the pairwise distances reflect closeness along the chosen category axis.
- Extensive experiments demonstrate that our lightweight transformations achieve performance matching or surpassing existing approaches across multiple attribute control tasks with minimal computational cost. A single hypernetwork is trained across **all** attributes/datasets simultaneously, for a total coverage of **19** different attributes with **187** unique subclasses.

2 Related Work

Foundation embedding models. Foundation models for vision-text embedding have demonstrated that high-dimensional representations can capture rich semantic relationships from large-scale text data [34, 38]. This principle has been extended to multimodal contexts by models such as CLIP [39], ALIGN [15], and SigLIP [66], which learn a joint embedding space for images and text. Subsequent work has further scaled and refined such embeddings to improve representation quality and generalization [20, 36, 47, 51]. Our approach operates on these frozen embeddings, allowing seamless integration into existing pipelines without recomputing features, and avoids fine-tuning the encoder, thereby mitigating forgetting or performance degradation.

Powerful multimodal embeddings have enabled applications such as semantic image retrieval [5, 35, 43, 67] and zero-shot classification [32, 39, 61, 68]. Many works proposed lightweight task adaptations of such models to improve downstream performance [13, 16]. Composed or conditional image retrieval methods extend these capabilities by allowing a user-specified text modification to retrieve images along different semantic axes [1–3, 11, 27, 28, 45, 52, 54, 56]. In contrast, our method focuses on *controlling embeddings* according to a text query, either by transforming the query alone for retrieval or by transforming the full embedding space, rather than retrieving a specific modified image.

Attribute disentanglement/attribute-aware embeddings. While foundation models like CLIP and ALIGN provide rich multimodal embeddings, fine-grained attributes such as color, pose, and lighting are often entangled with higher-level semantics [46]. Recent approaches have sought to disentangle visual embeddings into separate attribute and object components. For example, [12, 41] learn disentangled representations to support compositional reasoning and fashion-based retrieval, respectively, but require retraining, which can be computationally expensive. Other works adopt lightweight approaches using frozen embeddings [48, 60], but focus primarily on personalization for text-to-image generation. Another line of research aims to learn interpretable attribute bits for hash-code-based retrieval [6, 7, 30, 42, 55, 58, 59]. In contrast, our method proposes a hypernetwork to generate controlled transformations of embeddings, enabling robust disentanglement without retraining the base encoder.

Multi-clustering methods. Multi-clustering aims to cluster the same dataset in various meaningful ways. Traditional methods often require careful feature selection and design [14, 33, 40, 50, 57, 63]. Recent works like Multi-Map [65] and Multi-Sub [64] leverage powerful pretrained models like CLIP but fine-tune the model on each user query, optimizing clustering-specific losses that can disturb the natural relational structure. Another branch of methods utilizes large multimodal models (LMMs) such as LLaVA [24] to caption images, then employs large language models (LLMs) for iterative clustering based on text-based user inputs [21, 26, 31]. While these methods avoid retraining, they introduce significant inference costs. In contrast, our method learns to generate a text-conditioned projection that operates on frozen encoder features, offering a scalable and efficient solution for multi-clustering tasks.

3 Method

The main objective of this work is to construct meaningful, text-conditioned transformations of foundation model embedding spaces. Our approach learns a *single* hypernetwork that maps a natural language input (e.g., “color” or “camera angle”) to the parameters of an affine transformation, reshaping the embedding space to emphasize the queried attribute. This transformation can be applied either to (1) the retrieval query only for efficient attribute-aware retrieval without changing the pre-extracted visual features for the whole gallery, or (2) the full embedding set for global tasks such as clustering.

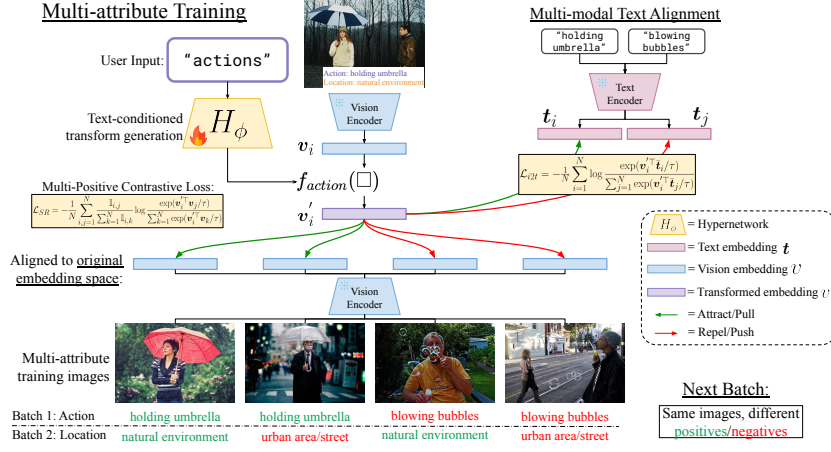


Fig. 2: Overview of our multi-attribute training workflow. An attribute category is sampled (e.g., “action” or “location”) and passed through the proposed hypernetwork to generate a transformation function, which is applied to a query embedding to produce attribute-focused representations aligned with base embeddings of images sharing the same category attribute label via a multi-positive contrastive loss. An additional multimodal guidance loss further aligns transformed features with the text embedding of the corresponding attribute label. The process iterates over the full set of valid attributes, where the set of positive/negative samples varies based on selected category.

3.1 Notation

We denote the vision and text encoders as f_v and f_t , respectively. Given an image I and a text input T , the encoders produce modality-specific embeddings:

$$\mathbf{v} = f_v(I), \quad \mathbf{t} = f_t(T),$$

where $\mathbf{v}, \mathbf{t} \in \mathbb{R}^D$. Unless otherwise specified, we use the CLIP ViT-L/14 model [39] with frozen weights. All operations are performed directly in the embedding space, eliminating the need for finetuning the base encoders or feature re-extraction.

We consider a dataset $\mathcal{D} = \{(I_i, \mathbf{y}_i)\}_{i=1}^N$, where each image I_i is annotated with labels $\mathbf{y}_i$ across M semantic taxonomies. Each category $\mathcal{C}^{(m)}$ (e.g., “color”) defines a set of K_m mutually exclusive classes $\{c_1^{(m)}, \dots, c_{K_m}^{(m)}\}$.

3.2 Text-Conditioned Affine Transformation Hypernetwork

We propose a hypernetwork H_ϕ that generates an affine transformation conditioned on a textual query. Given a category text embedding $\mathbf{t}$, the hypernetwork outputs the parameters of an affine transformation:

$$\mathbf{W}, \mathbf{b} = H_\phi(\mathbf{t}), \quad (1)$$

where $\mathbf{W} \in \mathbb{R}^{D \times D}$ and $\mathbf{b} \in \mathbb{R}^D$. Each image embedding $\mathbf{v}$ is then projected as:

$$\mathbf{v}' = \mathbf{v}\mathbf{W}^T + \mathbf{b}. \quad (2)$$

This can be applied to a single query embedding for attribute-aware retrieval in the *query* setting, or to the full embedding set $\mathbf{V}$ in the *global* setting to create a new embedding space $\mathbf{V}'$ specialized to the input attribute. Note that the conditioning signal does not fundamentally require text embeddings, it is chosen as a natural interface choice for specifying attributes.

Query-based Training Objective. The hypernetwork is trained to generate transformations that reorganize embeddings based on a user-specified category while maintaining consistency with the foundation model’s original embedding structure. An attribute category is sampled for each batch, defining the label set for the loss objectives. Diverse sampling results in a single hypernetwork capable of optimizing for multiple attributes simultaneously. We use a multi-positive contrastive loss based on StableRep [44] to encourage transformed samples to be near the base embeddings from the same subclass:

$$\mathcal{L}_{SR} = -\frac{1}{N} \sum_{i,j=1}^N \frac{\mathbb{1}_{i,j}}{\sum_{k=1}^N \mathbb{1}_{i,k}} \log \frac{\exp(\mathbf{v}'_i{}^\top \mathbf{v}_j / \tau)}{\sum_{k=1}^N \exp(\mathbf{v}'_i{}^\top \mathbf{v}_k / \tau)}, \quad (3)$$

where $\mathbf{v}'_i \in \mathbf{V}'$, $\mathbf{v}_j \in \mathbf{V}$ are ℓ_2 -normalized feature embeddings, τ is a temperature parameter, N is the batch size, and $\mathbb{1}_{i,j} = 1$ if i and j share a class label under the sampled attribute category and 0 otherwise.

This objective encourages each transformed embedding $\mathbf{v}'_i$ to align closely with embeddings $\mathbf{v}_j$ from the same attribute label within the base space. The query-only similarity score $\langle \mathbf{W}\mathbf{v}'_i + \mathbf{b}, \mathbf{v} \rangle$ factors as $\mathbf{v}'_i{}^\top \mathbf{W}^\top \mathbf{v} + \mathbf{b}^\top \mathbf{v}$. The product $\mathbf{v}'_i{}^\top \mathbf{W}^\top \mathbf{v}$ shows that the transformation is applied to the gallery side $\mathbf{v}$ via the inner product at score time rather than rewritten into the stored gallery vectors. Thus, the gallery can remain fixed while the retrieval score is axis-aware.

Multimodal Objective. To maintain cross-modal consistency, we also include a symmetric multimodal alignment loss between transformed visual embeddings and text embeddings of the corresponding attribute labels:

$$\mathcal{L}_{i2t} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\mathbf{v}'_i{}^\top \bar{\mathbf{t}}_i / \tau)}{\sum_{j=1}^N \exp(\mathbf{v}'_i{}^\top \bar{\mathbf{t}}_j / \tau)}, \quad (4)$$

where $\bar{\mathbf{t}}_i$ represents an averaged text embedding over descriptive templates (e.g., “a photo of {c}”, “an image in the {c} style”). $\mathcal{L}_{t2i}$ is defined in a similar fashion, just with $\mathbf{v}'$ swapped with $\bar{\mathbf{t}}$.

The final loss combines the multi-positive visual term and the multimodal guidance terms:

$$\mathcal{L} = \mathcal{L}_{SR} + \omega * (\mathcal{L}_{i2t} + \mathcal{L}_{t2i}), \quad (5)$$

where ω balances text alignment and category structure.

Dual Inference Capability. Although the hypernetwork is trained only in the *query* setting, where transformed embeddings $\mathbf{v}'$ are aligned to the frozen base

space, it can be used for a dual purpose during inference time. The first straightforward application is to transform the query image, yielding an attribute-aware retrieval along the direction of the transformation. The second ability is to achieve an emergent *global* transformation that reorganizes the entire distribution along the specified attribute axis without additional training. This transformed distribution shapes clustering to identify meaningful groupings based on the provided attribute. Concretely, the same learned transformation can be applied directly to the entire embedding set ($\mathbf{V} \rightarrow \mathbf{V}'$) at inference time. To summarize, the model learns a general operator that adapts to any visual embedding collection in an attribute-aware manner, enabling both attribute-aware retrieval and multi-attribute clustering within a single unified framework.

Table 1: Frozen gallery attribute-aware retrieval results reported as mAP averaged across dataset taxonomies. Our method simultaneously handles multiple categories with one set of weights while drastically improving performance. † denotes MLP adapter trained with multi-positive loss.

Method	Multi-Attr. Capability	Clevr4 mAP	S40 mAP	ShotBench mAP
Base Features	✗	33.8	43.7	25.2
LCR	✓	15.9	19.9	18.9
CWP	✓	43.8	49.8	21.4
ArcFace [8]†	✗	12.3	18.8	17.5
SupCon [17]†	✗	12.7	18.8	17.7
StableRep+ [44]†	✗	49.5	21.3	20.9
MaPLe [16]	✗	33.5	47.8	23.7
LoRA [13]	✗	42.9	49.3	22.0
SEARLE-XL [2]	✗	24.6	33.9	20.3
Ours-Query	✓	77.0	86.3	41.0

Table 2: Multi-clustering results averaged over 14 attribute categories. Inference time is reported assuming a small evaluation set of 5k images. The proposed approach improves performance while offering minimal compute overhead and near-instant inference. (*reported on full action set.)

Method	Inference Time (s)	Clevr4 c. ACC	S40 c. ACC	ShotBench c. ACC
Base Features	0.59	48.8	65.2	20.8
LCR	0.62	56.4	64.8	30.2
CWP	0.68	60.3	64.9	32.2
Multi-MAP [65]	750	62.8	62.8	32.6
Multi-Sub [64]	15300	72.2	66.1	32.4
IC TC [21]	37680	57.9	76.1	21.7
SSD-LLM [31]	27600	56.5	74.1	25.0
X-Cluster [26]	104760	64.9	68.3*	-
SEARLE-XL [2]	81	30.7	46.9	24.5
Ours-Space	0.64	73.0	80.0	51.6

4 Experiments

We first describe the experimental setup, then present results on attribute-aware retrieval, clustering, and show comprehensive ablation studies and analysis. Additional experimental results can be found in [Supplementary Section C](#).

4.1 Datasets

ShotBench: ShotQA/ShotBench [25] are datasets designed for training and evaluating vision-language models on question-answering related to fine-grained photographic attributes, namely camera angle, composition, lens size, lighting, lighting type, shot framing, and shot size. From these, we extract attribute answers as labels, resulting in a multi-label dataset consisting of 31k training images and 2.7k testing images across 7 categories.

Clevr-4 [49] is a synthetic dataset of 10k image samples, where each image contains a set of shapes with 4 axes of variation: color, count, shape, and texture. Each attribute category contains 10 subclasses and a mainly even distribution, so no subclass is over/underrepresented.

Stanford 40 Actions [62] contains 9,532 images of humans performing one of 40 actions, along with attribute annotations for 1,000 images with 10 location categories and 4 mood categories [21], resulting in three distinct categories: action, location, and mood. For training, we use a subset of 532 images drawn from the original action training split, annotating them with labels consistent with the existing taxonomy. Evaluation follows the same setup as prior work, using the full mood and location subsets and the remaining 9,000 images for action.

MS-COCO [23]. We additionally use the COCO 2017 test split (41k images) for qualitative examples, serving as a diverse, real-world benchmark to illustrate cross-domain generalization of attribute transformations learned on synthetic or domain-specific datasets.

Additional information on datasets used may be found in [Supp. Section A](#).

4.2 Baselines

Few existing methods allow controllable adaptation of frozen embedding spaces, so we construct two lightweight, *training-free* baselines to contextualize the benefits of our learned transformation. These baselines operate entirely using CLIP text and image embeddings and provide inexpensive, interpretable attribute-specific projections.

Given an attribute category with K class labels, we form a concept matrix $\mathbf{A} = [\mathbf{t}_1, \dots, \mathbf{t}_K] \in \mathbb{R}^{K \times D}$, where each row $\mathbf{t}_k$ is the normalized text embedding of one attribute label. Both baselines project image embeddings $\mathbf{V} \in \mathbb{R}^{N \times D}$ into the subspace spanned by these concept vectors.

(1) Linear Concept Reconstruction (LCR). A simple reconstruction of each visual embedding in the concept span:

$$\mathbf{V}_{\text{LCR}} = (\mathbf{V}\mathbf{A}^T)\mathbf{A}. \quad (6)$$

LCR treats the attribute directions as a basis and re-expresses each image embedding as a weighted combination of them.

(2) Concept-Weighted Projection (CWP). Because the concept vectors are not orthogonal, we correct for overlap via the inverse Gram matrix $(\mathbf{A}\mathbf{A}^T)^{-1}$:

$$\mathbf{V}_{\text{CWP}} = (\mathbf{V}\mathbf{A}^T)(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A}. \quad (7)$$

This performs the closest-point projection onto the attribute subspace, effectively removing redundancy between overlapping concept directions.

Both baselines offer parameter-free, low-cost attribute awareness, but lack the learnable expressive ability of our proposed text-conditioned hypernetwork.

4.3 Implementation Details

All experiments are implemented in PyTorch [37] and trained on a single NVIDIA A100 GPU. The hypernetwork is a three-layer MLP with ReLU activations, which takes as input the CLIP text embedding and predicts the parameters of the affine transformation $(\mathbf{W}, \mathbf{b})$. Dimensions depend on the chosen CLIP variant (e.g., 768 for ViT-L/14). We train with the AdamW [18, 29] optimizer, a learning rate of $1e-4$, and multimodal loss weight $\omega = 0.1$. All reported runs use frozen encoder weights; only the hypernetwork parameters are updated during training. Further implementation details are provided in [Supp. Section B](#).

4.4 Attribute-aware Image-to-Image Retrieval

We begin with the attribute-aware retrieval setting, where only the query embedding is transformed while retrieval is performed over the original frozen gallery. This *query* mode preserves compatibility with large-scale search systems, as no re-encoding per attribute is required. Table 1 shows that CLIP’s base features do not offer control and struggle to surface attribute-consistent matches. Lightweight adapter-based approaches such as LoRA [13] and MaPLe [16] fine-tune CLIP to generally improve alignment. However, these require separate sets of parameters per attribute category, causing memory and storage costs to scale linearly with attribute count, and fail to handle the frozen gallery. In contrast, our method achieves state-of-the-art performance in this setting, while natively handling multiple categories and keeping memory footprint constant: a single hypernetwork handles all categories jointly, without growing in size as the number of attributes increases. Expanded results are provided in [Supp. Section C](#).

4.5 Unsupervised Multi-Clustering

We next evaluate the *global* setting, where the learned transformation is applied to the entire embedding set and unsupervised k -means clustering is performed on the transformed features. For each attribute, we run Faiss k -means [9] and compute clustering accuracy using Hungarian matching [19]. Table 2 compares our approach with state-of-the-art multi-clustering methods including Multi-MaP [65], Multi-Sub [64], IC|TC [21], SSD-LLM [31], and X-Cluster [26]. Methods that rely on image captioning and LLM iteration [21, 26, 31] incur high inference cost because each image must be captioned and repeatedly queried through a large language model, which are operations that scale linearly with dataset size and dominate runtime. [64, 65] reduce this reliance but still require per-attribute finetuning after user input, yielding moderate performance but still with substantial compute and memory overhead. In contrast, our proposed query-focused training method naturally handles this attribute disentanglement setting and achieves the highest average clustering accuracy across datasets while using a *single* transformation network trained once for all attributes.



Fig. 3: Qualitative attribute-aware retrieval results under different attribute transformation functions. Our text-conditioned query transformations selectively retrieve images that match the specified attribute without losing the scene semantics at first, then prioritize the attribute over general semantics. From top to bottom, we show retrievals for "action", "count", then fine-grained perceptual property "camera angle".

4.6 Ablations and Analysis

Transformation Type. Attributes are entangled in CLIP space, yet they remain linearly separable [4]. We demonstrate that controllable similarity is achievable with an asymmetric affine transformation applied to the query vector alone. A simple offset vector (translation) does not have the ability to scale and rotate sub-spaces, which is needed to satisfy our setting constraints (Table 3).

Table 3: Affine transforms effectively enable per-attribute similarity.

Dataset	Variant	Cluster ACC	Query mAP	Global mAP
Clevr-4	Base	32.2	33.8	—
	Translation-only	32.2	35.4	34.5
	Affine	75.4 (+43.2)	77.6 (+42.2)	85.1 (+50.6)
Stanford-40	Base	46.4	43.6	—
	Translation-only	55.6	43.8	44.5
	Affine	79.4 (+23.8)	84.4 (+40.6)	87.7 (+43.2)

Compositional Retrieval. Our framework directly supports compositional retrieval without retraining. We compose a multi-attribute query by linearly blending the projected queries into $q_{\text{blend}} = \alpha (W_a q + b_a) + (1 - \alpha) (W_b q + b_b)$, where a and b are attributes such as action and location. Table 4 shows that a composed query ($\alpha = 0.5$) improves joint-label (Both) retrieval over the raw baseline and either single-axis projection, indicating that the learned projections can be combined for multi-attribute user queries.

Table 4: Compositional retrieval on Stanford40 action (A) + location (L). Uniform blending improves joint-label retrieval w/o retraining.

Variant	mAP(Both)	Both@5	A-only@5	L-only@5	Neither@5
Base	48.5	45.1	21.5	15.9	17.5
Action only	70.0	57.4	40.7	0.9	1.0
Location only	21.2	15.6	2.8	76.4	5.3
Linear Blend ($\alpha = 0.5$)	80.9	64.8	30.2	3.6	1.4

Embedding Space Analysis. Our trained transformation $\mathbf{W}$ keeps the directions in the embedding that carry the target attribute information and suppresses the rest, so that components orthogonal to the “kept” directions do not contribute to the cosine similarity. Attribute-matched images then rank higher than mismatched ones, even for a frozen gallery. We provide two direct findings: **1) The transformed feature space is lower-dimensional and attribute-specific.** Stable rank analysis (Tab. 5) shows that projected features $\mathbf{v}_i \cdot \mathbf{W}$ collapses, with simpler attributes (color) showing a bigger collapse. Additionally, the column space of $\mathbf{W}$ carries the attribute signal; projecting features into it and running retrieval greatly exceeds the raw baseline, while features projected orthogonal to $\text{col}(\mathbf{W})$ collapses. This strongly supports our hypothesis of an operator selecting a small attribute-specific subspace.

2) Query-only projection improves score discriminability. Cosine similarities drop in absolute magnitude after projection (Tab. 6, transformed vs. base-gallery), but the gap between matched and unmatched widens, and ranking AUC also rises sharply in that axis. This is consistent with our hypothesis that $\mathbf{W}$ “suppresses irrelevant components”.

Table 5: Stable rank and retrieval analysis. “Col. space” projects onto the column space of $\mathbf{W}$. “Orth. Comp.” indicates the orthogonal complement. The transform operation selects a small attribute-specific subspace.

Axis	Stable Rank (base→trans)	Base mAP	Col. Space	Orth. Comp.
Color	11.3 → 5.0	11.9	98.4	4.9
Action	22.1 → 14.3	48.9	92.1	15.3

Table 6: Average cosine similarities for same/diff.-label pairs on Clevr4. AUC measures whether same-label pairs rank above different-label pairs. Transformed features drastically improve inter-class separation.

Axis	Mode	Same/Diff.	Cohen’s d	AUC
Color	Raw	0.76 / 0.72	0.7	0.703
	Query	0.19 / -0.02	4.0	0.993
Count	Raw	0.74 / 0.72	0.3	0.579
	Query	0.10 / 0.01	1.4	0.829

Generalization to Unseen Classes. Table 7 evaluates color-conditioned retrieval when specific color subclasses are held out during training. Our transformed embeddings still retrieve images of the same color as the query, despite not seeing it during training, indicating that the transformation successfully emphasizes the general color property instead of only learning the specific seen colors. However, similar colors (blue/cyan) struggle to fully disentangle.

Table 7: Color generalization evaluation on Clevr4. Ours (seen) is the upper bound where all colors are present during training. Ours (unseen) indicates training with one color held out (e.g., no green), then evaluating color-based retrieval (mAP) on that color. Our transformation functions learn to generalize beyond the training subclasses.

Method	Clevr4 – Color Retrieval (mAP)										Avg.
	Blue	Brown	Cyan	Gray	Green	Orange	Pink	Purple	Red	Yellow	
Ours (seen)	99.6	93.9	99.6	99.3	99.6	86.2	88.5	99.6	98.4	74.2	93.9
Baseline	23.3	16.9	26.5	21.0	28.6	21.0	20.4	30.1	21.5	19.4	22.9
Ours (unseen)	33.1	74.5	32.6	47.4	91.9	84.2	44.4	90.2	42.3	39.5	58.0
	+9.8	+57.6	+6.1	+26.4	+63.3	+63.2	+24.0	+60.1	+20.8	+20.1	+35.1

Effect of Multimodal Guidance. Beyond improving retrieval and clustering, multimodal alignment also enhances text-similarity classification accuracy (Table 8), reflecting tighter correspondence between transformed embeddings and attribute text features. Even without explicit supervision, the transformation exhibits meaningful alignment due to CLIP’s pretrained cross-modal structure.

Table 8: Ablation on multimodal guidance across three tasks using native metrics. The multimodal objective (Eq. 4) enables text-similarity classification and modestly improves retrieval and clustering performance.

Objective	Multiple Clustering			Attr.-Aware Retrieval			Text-based Classification		
	Clevr4	S40	ShotBench	Clevr4	S40	ShotBench	Clevr4	S40	ShotBench
Ours	73.0	80.0	51.6	77.1	91.2	41.1	88.7	83.2	51.6
– multimodal	72.7	78.4	53.2	76.8	87.3	40.7	71.7	78.5	31.3

Data Efficiency. Figure 4 ablates the number of training samples used per class on the Stanford 40 Action [62] dataset. Remarkably, our global affine transformation achieves near peak performance across all taxonomies given just **six** training samples per class.

Scalability Across Attributes. We assess how well the model scales as more attribute categories are trained within a single hypernetwork. Starting from one model per attribute, we progressively expand to one per dataset and finally to a unified model trained across all datasets. The results in Figure 5 show that performance remains stable even as the number of jointly trained attributes increases, varying by less than 1.5%. Notably, final results are presented using a single hypernetwork trained across **19** attributes with **187** subclasses. This consistency indicates that the shared text-conditioned transformation can generalize across many attributes without interference, enabling efficient multi-attribute learning within a single network.

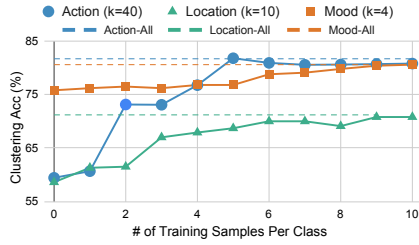


Fig. 4: Multi-clustering accuracy on Stanford40 under varying number of training images per class. Category-All line indicates performance using all available training samples. Our transform approach demonstrates extreme data efficiency, achieving near peak results with just **six** images per class.

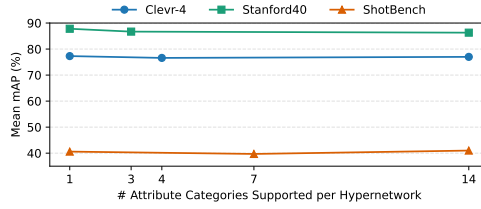


Fig. 5: Mean category mAP across datasets (each line) as the number of jointly trained attribute categories increases from a single category per model to unified training across all datasets. Performance remains consistent even as more attributes are added, demonstrating that a single hypernetwork can scale across 14+ attributes without degradation.

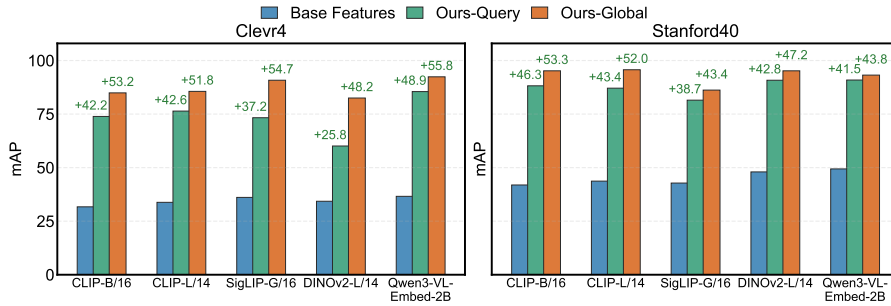


Fig. 6: Ablation on backbone architecture, evaluated using retrieval mAP. Our method is effective across all types/sizes of encoders, including non-text aligned DINOv2.

Backbone Encoder Choice. Figure 6 shows that the proposed transformation consistently improves attribute-based retrieval across encoder architectures and model sizes. Gains are observed for both vision-language models (CLIP, SigLIP, Qwen3-VL-Embed) and purely visual backbones (DINOv2), demonstrating that the method generalizes beyond text-aligned representations. Note that for DINOv2, the CLIP H/14 text encoder is used for consistent feature dimensionality.

4.7 Discussion

We next analyze qualitative and emergent properties of learned transformations. **How can a single transformed query embedding retrieve from a frozen gallery set?** The transformation is trained with a multi-positive contrastive loss which optimizes relative similarity—pulling the transformed query closer to positives while repelling negatives. Because most concepts are partially entangled (e.g., two images may share an object but differ in color), minimizing absolute

distance to positives would inadvertently bring it near some negatives. To resolve this, the network learns to move the query in a new direction within the high-dimensional space that separates it from the general visual manifold while aligning it with the target attribute. This broadly resembles the text–image “modality gap” [22], allowing the transformed query to behave similar to a text embedding. Even without explicit multimodal supervision, Table 8 shows that transformed queries naturally align with text features, enabling accurate retrieval from a frozen gallery using similarity in the new space.

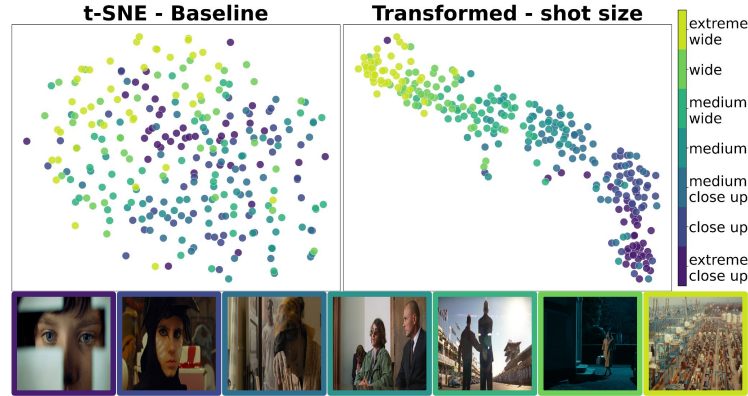


Fig. 7: t-SNE visualization of embeddings colored by shot-size labels before and after transformation. The baseline space (left) shows entangled clusters. After applying our learned transformation (right), samples align intuitively along a continuous gradient that reflects the shot-size taxonomy.

Can the transformation capture perceptual attributes and how does it affect embedding geometry? Contrastive image–text models such as CLIP are trained to capture broad semantic meaning, yet they provide little explicit separation for perceptual attributes like shot size, composition, or camera angle. As a result, embeddings that differ in these subtle properties often remain close in the feature space. Our transformation isolates these attributes by reorienting the space toward the specified taxonomy while preserving global structure. As shown in Figure 7, applying the transformation introduces a clear, continuous axis of variation corresponding to the target attribute (here, shot size). Samples form a smooth progression from close-up to wide shots, reflecting a structured and interpretable organization. Quantitatively, this behavior yields strong improvements on fine-grained perceptual benchmarks like ShotBench, confirming that the transformation reshapes latent geometry to make attribute-specific relationships explicit without distorting the original feature layout.

Do attribute-specific transforms learned on synthetic data transfer to real images? Our transformations generalize cleanly across domains. When trained on the synthetic Clevr-4 dataset, which contains simple geometric ob-

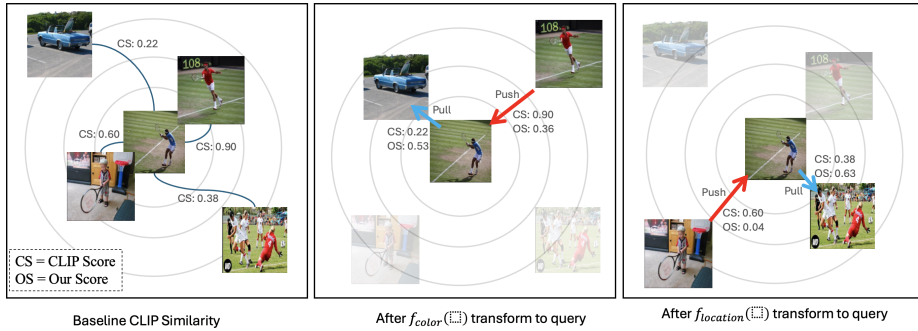


Fig. 8: Our attribute-conditioned similarity metric helps characterize unique relationships that CLIP similarity alone cannot. (Left) Baseline CLIP similarity prioritizes overall semantics. (Middle, Right) Applying attribute-specific transforms to the query (e.g., color or location) selectively re-scores relationships based on the input attribute.

jects with uniform backgrounds, the model still produces meaningful attribute disentanglement for real, diverse MS-COCO images (Figs. 3 and 8). Despite the distribution shift, transforms learned for color and count correctly align COCO embeddings along the intended dimensions. We conjecture that this generalization arises from optimizing a single affine transform for the entire embedding space, which discourages shortcut memorization and instead forces the transform to properly disentangle embeddings to optimize the contrastive loss.

Attribute-aware similarity. Figure 8 illustrates how the proposed similarity metric handles relationships that are not captured by standard CLIP similarity. Images that share the attribute are pulled closer while others are pushed away, irrespective of their original CLIP similarity. Our transform effectively restructures the embedding space to emphasize the selected attribute, enabling controlled comparison of images along different attribute dimensions.

Composed Image Retrieval (CIR) Compatibility. Our transformations can also be integrated into existing CIR pipelines to improve attribute discriminability. We provide an illustrative experiment in [Supp. Section C](#).

5 Conclusion

We presented a text-conditioned transformation framework that enables fine-grained control over pretrained embedding spaces without modifying encoder weights or recomputing gallery features. A single hypernetwork learns to generate attribute-specific affine transformations from natural-language descriptions, supporting both attribute-aware retrieval and multi-attribute organization with near-zero inference overhead. While the method generalizes across diverse attributes, its expressiveness is currently bounded by the range of categories seen during training. Future work may explore scaling up training to find emergent transformation properties across attributes completely unseen during training.

References

1. Agnolucci, L., Baldrati, A., Del Bimbo, A., Bertini, M.: isearle: Improving textual inversion for zero-shot composed image retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025)
2. Baldrati, A., Agnolucci, L., Bertini, M., Del Bimbo, A.: Zero-shot composed image retrieval with textual inversion. In: *Proceedings of the IEEE/CVF international conference on computer vision*. pp. 15338–15347 (2023)
3. Baldrati, A., Bertini, M., Uricchio, T., Del Bimbo, A.: Conditioned and composed image retrieval combining and partially fine-tuning clip-based features. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 4959–4968 (2022)
4. Bhalla, U., Oesterling, A., Srinivas, S., Calmon, F.P., Lakkaraju, H.: Interpreting clip with sparse linear concept embeddings (splice). *Advances in Neural Information Processing Systems* **37**, 84298–84328 (2024)
5. Cao, B., Araujo, A., Sim, J.: Unifying deep local and global features for image search. In: *European conference on computer vision*. pp. 726–743. Springer (2020)
6. Chen, Z.D., Zhao, L.J., Zhang, Z.C., Luo, X., Xu, X.S.: Characteristics matching based hash codes generation for efficient fine-grained image retrieval. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 17273–17281 (2024)
7. Cui, Q., Jiang, Q.Y., Wei, X.S., Li, W.J., Yoshie, O.: Exchnet: A unified hashing network for large-scale fine-grained image retrieval. In: *European Conference on Computer Vision*. pp. 189–205. Springer (2020)
8. Deng, J., Guo, J., Xue, N., Zafeiriou, S.: Arcface: Additive angular margin loss for deep face recognition. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 4690–4699 (2019)
9. Douze, M., Guzhva, A., Deng, C., Johnson, J., Szilvasy, G., Mazaré, P.E., Lomeli, M., Hosseini, L., Jégou, H.: The faiss library (2024)
10. Gao, P., Geng, S., Zhang, R., Ma, T., Fang, R., Zhang, Y., Li, H., Qiao, Y.: Clip-adapter: Better vision-language models with feature adapters. *International Journal of Computer Vision* **132**(2), 581–595 (2024)
11. Gupta, A., Parmar, J., Dave, I.R., Shah, M.: From play to replay: Composed video retrieval for temporally fine-grained videos. *arXiv preprint arXiv:2506.05274* (2025)
12. Hou, Y., Vig, E., Donoser, M., Bazzani, L.: Learning attribute-driven disentangled representations for interactive fashion retrieval. In: *Proceedings of the IEEE/CVF International conference on computer vision*. pp. 12147–12157 (2021)
13. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al.: Lora: Low-rank adaptation of large language models. *ICLR* **1**(2), 3 (2022)
14. Hu, J., Qian, Q., Pei, J., Jin, R., Zhu, S.: Finding multiple stable clusterings. *Knowledge and Information Systems* **51**(3), 991–1021 (2017)
15. Jia, C., Yang, Y., Xia, Y., Chen, Y.T., Parekh, Z., Pham, H., Le, Q., Sung, Y.H., Li, Z., Duerig, T.: Scaling up visual and vision-language representation learning with noisy text supervision. In: *International conference on machine learning*. pp. 4904–4916. PMLR (2021)
16. khattak, M.U., Rasheed, H., Maaz, M., Khan, S., Khan, F.S.: Maple: Multi-modal prompt learning. In: *The IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2023)
17. Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., Krishnan, D.: Supervised contrastive learning. *Advances in neural information processing systems* **33**, 18661–18673 (2020)

18. Kingma, D.P.: Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980 (2014)
19. Kuhn, H.W.: The hungarian method for the assignment problem. *Naval research logistics quarterly* **2**(1-2), 83–97 (1955)
20. Kusupati, A., Bhatt, G., Rege, A., Wallingford, M., Sinha, A., Ramanujan, V., Howard-Snyder, W., Chen, K., Kakade, S., Jain, P., et al.: Matryoshka representation learning. *Advances in Neural Information Processing Systems* **35**, 30233–30249 (2022)
21. Kwon, S., Park, J., Kim, M., Cho, J., Ryu, E.K., Lee, K.: Image clustering conditioned on text criteria. arXiv preprint arXiv:2310.18297 (2023)
22. Liang, V.W., Zhang, Y., Kwon, Y., Yeung, S., Zou, J.Y.: Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. *Advances in Neural Information Processing Systems* **35**, 17612–17625 (2022)
23. Lin, T.Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft coco: Common objects in context. In: *European conference on computer vision*. pp. 740–755. Springer (2014)
24. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. *Advances in neural information processing systems* **36**, 34892–34916 (2023)
25. Liu, H., He, J., Jin, Y., Zheng, D., Dong, Y., Zhang, F., Huang, Z., He, Y., Li, Y., Chen, W., Qiao, Y., Ouyang, W., Zhao, S., Liu, Z.: Shotbench: Expert-level cinematic understanding in vision-language models (2025), <https://arxiv.org/abs/2506.21356>
26. Liu, M., Zhong, Z., Li, J., Franchi, G., Roy, S., Ricci, E.: Organizing unstructured image collections using natural language. arXiv preprint arXiv:2410.05217 (2025)
27. Liu, Z., Rodriguez-Opazo, C., Teney, D., Gould, S.: Image retrieval on real-life images with pre-trained vision-and-language models. In: *Proceedings of the IEEE/CVF international conference on computer vision*. pp. 2125–2134 (2021)
28. Liu, Z., Sun, W., Teney, D., Gould, S.: Candidate set re-ranking for composed image retrieval with dual multi-modal encoder. arXiv preprint arXiv:2305.16304 (2023)
29. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. arXiv preprint arXiv:1711.05101 (2017)
30. Lu, X., Chen, S., Cao, Y., Zhou, X., Lu, X.: Attributes grouping and mining hashing for fine-grained image retrieval. In: *Proceedings of the 31st ACM International Conference on Multimedia*. pp. 6558–6566 (2023)
31. Luo, Y., An, R., Zou, B., Tang, Y., Liu, J., Zhang, S.: Llm as dataset analyst: Subpopulation structure discovery with large language model. In: *European Conference on Computer Vision*. pp. 235–252. Springer (2025)
32. Menon, S., Vondrick, C.: Visual classification via description from large language models. arXiv preprint arXiv:2210.07183 (2022)
33. Miklautz, L., Mautz, D., Altinigneli, M.C., Böhm, C., Plant, C.: Deep embedded non-redundant clustering. In: *Proceedings of the AAAI conference on artificial intelligence*. vol. 34, pp. 5174–5181 (2020)
34. Mikolov, T., Chen, K., Corrado, G., Dean, J.: Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781 (2013)
35. Noh, H., Araujo, A., Sim, J., Weyand, T., Han, B.: Large-scale image retrieval with attentive deep local features. In: *Proceedings of the IEEE international conference on computer vision*. pp. 3456–3465 (2017)
36. Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al.: Dinov2: Learning robust visual features without supervision. arXiv preprint arXiv:2304.07193 (2023)

37. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chillemi, G., DePasquale, L., Gitman, T., Li, J., Lin, Z., et al.: Pytorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems* **32** (2019)
38. Pennington, J., Socher, R., Manning, C.D.: Glove: Global vectors for word representation. In: *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*. pp. 1532–1543 (2014)
39. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. In: *International conference on machine learning*. pp. 8748–8763. PmLR (2021)
40. Ren, L., Yu, G., Wang, J., Liu, L., Domeniconi, C., Zhang, X.: A diversified attention model for interpretable multiple clusterings. *IEEE Transactions on Knowledge and Data Engineering* **35**(9), 8852–8864 (2022)
41. Saini, N., Pham, K., Shrivastava, A.: Disentangling visual embeddings for attributes and objects. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 13658–13667 (2022)
42. Shen, Y., Sun, X., Wei, X.S., Jiang, Q.Y., Yang, J.: Semicon: A learning-to-hash solution for large-scale fine-grained image retrieval. In: *European conference on computer vision*. pp. 531–548. Springer (2022)
43. Siméoni, O., Avrithis, Y., Chum, O.: Local features and visual words emerge in activations. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 11651–11660 (2019)
44. Tian, Y., Fan, L., Isola, P., Chang, H., Krishnan, D.: Stablerep: Synthetic images from text-to-image models make strong visual representation learners. *Advances in Neural Information Processing Systems* **36**, 48382–48402 (2023)
45. Tian, Y., Newsam, S., Boakye, K.: Fashion image retrieval with text feedback by additive attention compositional learning. In: *Proceedings of the IEEE/CVF winter conference on applications of computer vision*. pp. 1011–1021 (2023)
46. Tong, S., Liu, Z., Zhai, Y., Ma, Y., LeCun, Y., Xie, S.: Eyes wide shut? exploring the visual shortcomings of multimodal llms. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 9568–9578 (2024)
47. Tschannen, M., Gritsenko, A., Wang, X., Naeem, M.F., Alabdulmohsin, I., Parthasarathy, N., Evans, T., Beyer, L., Xia, Y., Mustafa, B., et al.: Siglip 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786* (2025)
48. Valevski, D., Lumen, D., Matias, Y., Leviathan, Y.: Face0: Instantaneously conditioning a text-to-image model on a face. In: *SIGGRAPH Asia 2023 Conference Papers*. pp. 1–10 (2023)
49. Vaze, S., Vedaldi, A., Zisserman, A.: No representation rules them all in category discovery. *Advances in Neural Information Processing Systems* **37** (2023)
50. Veit, A., Belongie, S., Karaletsos, T.: Conditional similarity networks. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 830–838 (2017)
51. Venkataramanan, S., Pariza, V., Salehi, M., Knobel, L., Gidaris, S., Ramzi, E., Bursuc, A., Asano, Y.M.: Franca: Nested matryoshka clustering for scalable visual representation learning. *arXiv preprint arXiv:2507.14137* (2025)
52. Vo, N., Jiang, L., Sun, C., Murphy, K., Li, L.J., Fei-Fei, L., Hays, J.: Composing text and image for image retrieval-an empirical odyssey. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 6439–6448 (2019)

53. Wang, J., Xu, Y., Hu, J., Yan, M., Sang, J., Qian, Q.: Improved visual fine-tuning with natural language supervision. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 11899–11909 (2023)
54. Wang, L., Ao, W., Boddeti, V.N., Lim, S.N.: Generative zero-shot composed image retrieval. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 29690–29700 (2025)
55. Wang, P., Li, Y., Zhao, L., Wei, X.S.: Learning attribute-aware hash codes for fine-grained image retrieval via query optimization. arXiv preprint arXiv:2509.17049 (2025)
56. Wang, T., Zhao, Y., Kong, S.: Generating a paracosm for training-free zero-shot composed image retrieval. arXiv preprint arXiv:2602.00813 (2026)
57. Wei, S., Wang, J., Yu, G., Domeniconi, C., Zhang, X.: Multi-view multiple clusterings using deep matrix factorization. In: Proceedings of the AAAI conference on artificial intelligence. vol. 34, pp. 6348–6355 (2020)
58. Wei, X.S., Shen, Y., Sun, X., Wang, P., Peng, Y.: Attribute-aware deep hashing with self-consistency for large-scale fine-grained image retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **45**(11), 13904–13920 (2023)
59. Wei, X.S., Shen, Y., Sun, X., Ye, H.J., Yang, J.: A2-net: Learning attribute-aware hash codes for large-scale fine-grained image retrieval. *Advances in Neural Information Processing Systems* **34**, 5720–5730 (2021)
60. Wei, Y., Zhang, Y., Ji, Z., Bai, J., Zhang, L., Zuo, W.: Elite: Encoding visual concepts into textual embeddings for customized text-to-image generation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 15943–15953 (2023)
61. Wortsman, M., Ilharco, G., Kim, J.W., Li, M., Kornblith, S., Roelofs, R., Lopes, R.G., Hajishirzi, H., Farhadi, A., Namkoong, H., et al.: Robust fine-tuning of zero-shot models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 7959–7971 (2022)
62. Yao, B., Jiang, X., Khosla, A., Lin, A.L., Guibas, L., Fei-Fei, L.: Human action recognition by learning bases of action attributes and parts. In: 2011 International conference on computer vision. pp. 1331–1338. IEEE (2011)
63. Yao, J., Liu, E., Rashid, M., Hu, J.: Augdmc: Data augmentation guided deep multiple clustering. *Procedia Computer Science* **222**, 571–580 (2023)
64. Yao, J., Qian, Q., Hu, J.: Customized multiple clustering via multi-modal subspace proxy learning. *Advances in Neural Information Processing Systems* **37**, 82705–82725 (2024)
65. Yao, J., Qian, Q., Hu, J.: Multi-modal proxy learning towards personalized visual multiple clustering. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 14066–14075 (2024)
66. Zhai, X., Mustafa, B., Kolesnikov, A., Beyer, L.: Sigmoid loss for language image pre-training. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 11975–11986 (2023)
67. Zhang, Z., Wang, L., Zhou, L., Koniusz, P.: Learning spatial-context-aware global visual feature representation for instance image retrieval. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 11250–11259 (2023)
68. Zhou, K., Yang, J., Loy, C.C., Liu, Z.: Conditional prompt learning for vision-language models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 16816–16825 (2022)